Networks, Layering and the Application Layer

Exercises 1

Computer Networks and Distributed Programming

What this pack is, and how to use it

- Thirty exercises on everything up to the application layer: networks and switching, layering, applications, HTTP, mail, DNS.
- They are for home. Work through them after the lectures, in order or by topic; they are the kind of question the exam asks.
- Each has a number, E1.1 to E1.30: bring the number to class, and the ones that caused trouble get cleared up there.
- There are no solutions here. A result you cannot check against a number is a result you have to argue for.
- Some exercises need a terminal, Wireshark or a browser. What to install is on the last slide.

Networks and switching

E1.1 What is the difference between a host and an end system? List several kinds of end system. Is a Web server an end system?

E1.2 What advantage does a circuit-switched network have over a packet-switched one? In a circuit-switched network, what advantages does TDM have over FDM?

E1.3 Why do two ISPs at the same level of the hierarchy often peer with each other? How does an IXP earn its money?

Topologies, counted

E1.4: Eight hosts must be connected to one another.

(a) For a bus, a ring, a star around one switch and a full mesh, how many links are needed? For the bus, count the backbone and one drop cable per host.

(b) For each topology, find a single failure — one link or one device — that cuts at least one host off from the others. How many hosts does it cut off? Is there a topology where no single link failure cuts anybody off?

(c) With n hosts instead of eight, how does the number of links grow in each of the four cases?

Metcalfe, and statistical multiplexing

E1.5 Metcalfe's law says that the value of a network grows with the square of the number of its users. If each of n users sends one message to each of the others, how many messages are sent? Does the answer support the law? Compare it with the links of the full mesh in E1.4.

E1.6 A 10 Mbps link is shared by users who each need 200 kbps when transmitting, but transmit only 10% of the time.

(a) With circuit switching, how many users can be supported?

(b) With packet switching, what is the probability that a given user is transmitting?

(c) With 120 users, what is the probability that exactly n are transmitting at once? And that 51 or more are?

Directions, sizes and rates

E1.7: Consider these six situations: (1) a wireless keyboard sends keystrokes to a PC; (2) two people talk on walkie-talkies; (3) a phone call over a fibre line; (4) the network joining the buildings of one university spread across a city; (5) a laptop plugged into the office switch; (6) two data centres on different continents.

(a) For (1), (2) and (3): simplex, half-duplex or full-duplex?

(b) For (1), (4), (5) and (6): PAN, LAN, MAN or WAN?

(c) A download shows 40 Mbit/s over a line sold as 100 Mbit/s. Which of the two figures is closer to the bandwidth of the path and which is the throughput? Give two reasons why they differ.

Layers

E1.8 Which layers of the Internet stack does a router process? A link-layer switch? A host?

E1.9 List five tasks that a layer can perform. Can the same task be performed by two or more layers? Give an example from the course.

Encapsulation, by hand

E1.10: A browser sends an HTTP request of 500 bytes over TCP, IPv4 and Ethernet. Assume a TCP header of 20 bytes, an IP header of 20 bytes, and an Ethernet header of 14 bytes plus a 4-byte check at the end of the frame.

(a) Draw the frame as nested boxes, each header next to what it wraps. Which layer adds something at the end as well?

(b) How many bytes does the adapter send, not counting any preamble? What fraction of them is the HTTP request?

(c) Repeat (b) for an application message of 10 bytes. What does the result say about sending many small messages?

(d) A router on the path receives this frame. Which headers does it read, and which does it rewrite before sending it on?

Applications

E1.11 For a P2P file-sharing application, do you agree that "there is no notion of client and server sides of a communication session"? Why?

E1.12 A transport protocol can offer an application four kinds of service: reliable delivery, throughput, timing and security. For each, say whether TCP, UDP, both or neither offers it.

Designing an application protocol

E1.13: Design the protocol between a cash machine and the bank's central computer. It must let the machine check a card and its PIN, ask for the balance, and withdraw money — including the common case in which the account does not hold enough.

(a) List the types of message, their syntax, the meaning of each field, and the rules for when each side sends what.

(b) Draw the exchange for a successful withdrawal as a time diagram, one line per side.

(c) State what your protocol assumes about the transport underneath. What happens if the machine's request arrives but the reply is lost — does the customer get the money twice, never, or once?

FTP, as it travels

E1.14: Start a capture, then open an anonymous session on a public FTP server (at the time of writing, `ftp.gnu.org` accepts the user `anonymous`):

```
$ ftp ftp.gnu.org          user:  anonymous, any password
ftp> ls
ftp> get README
ftp> quit
```

Display filter `ftp || ftp-data`.

(a) How many TCP connections were opened, towards which server ports, and which one stayed open for the whole session?

(b) Can you read the user name and the password? What follows?

(c) Which command and which reply set up each data connection, and who connected to whom? How does the client know the file is over?

Peer-to-peer distribution

E1.15 A new peer, Alice, joins a BitTorrent torrent holding no chunks. Having nothing to upload, she cannot become a top-four uploader for anybody. How, then, does she get her first chunk?

E1.16 A file of $F = 20\,\text{Gbit}$ must reach N peers. The server uploads at $u_s = 30\,\text{Mbps}$; each peer downloads at $d = 2\,\text{Mbps}$ and uploads at u. Using the minimum distribution times seen in the course, fill a table with one row per $N = 10$, 100, 1000 and one column per $u = 300\,\text{kbps}$, $700\,\text{kbps}$, $2\,\text{Mbps}$, for client-server and for P2P distribution. What does each architecture do as N grows?

HTTP: true or false

E1.17: True or false? Justify each answer in one line.

(a) A page made of some text and three images needs one request message and four response messages.

(b) `www.mit.edu/research.html` and `www.mit.edu/students.html` can be sent over the same persistent connection.

(c) With non-persistent connections, a single TCP segment can carry two distinct HTTP request messages.

(d) The `Date:` header of a response says when the object was last modified.

(e) HTTP responses never have an empty message body.

A request, as it was captured

E1.18: These are the exact bytes of a GET sent by a browser; <cr><lf> stands for the two characters carriage return and line feed.

```
GET /cs453/index.html HTTP/1.1<cr><lf>
Host: gaia.cs.umass.edu<cr><lf>
User-Agent: Mozilla/5.0 (Windows;U; Windows NT 5.1; en-US; rv:1.7.2) Gecko/20040804 Netscape
     /7.2<cr><lf>
Accept: text/xml, application/xml, application/xhtml+xml, text/html;q=0.9, */*;q=0.5<cr><lf>
Accept-Language: en-us,en;q=0.5<cr><lf>
Keep-Alive: 300<cr><lf>
Connection: keep-alive<cr><lf>
<cr><lf>
```

Say where in the message you find each answer. (a) The URL of the document. (b) The HTTP version. (c) Persistent or non-persistent connection? (d) The IP address of the browser's host. (e) The browser type — and why does a server want to know it?

The reply

E1.19: This is the server's reply to the request of E1.18.

```
HTTP/1.1 200 OK<cr><lf>
Date: Tue, 07 Mar 2008 12:39:45 GMT<cr><lf>
Server: Apache/2.0.52 (Fedora)<cr><lf>
Last-Modified: Sat, 10 Dec 2005 18:27:46 GMT<cr><lf>
ETag: "526c3-f22-a88a4c80"<cr><lf>
Content-Length: 3874<cr><lf>
Keep-Alive: timeout=max=100<cr><lf>
Connection: Keep-Alive<cr><lf>
Content-Type: text/html; charset=ISO-8859-1<cr><lf>
<cr><lf>
<!doctype html public "-//w3c//dtd html 4.0 transitional//en"><lf>
<html><lf> ... (the rest of the document)
```

(a) Was the document found, and when was the reply generated? (b) When was the document last modified? (c) How many bytes does it contain? (d) What are its first five bytes? Did the server agree to a persistent connection?

Counting round trips

E1.20: You click on a link. The IP address of the server is not in your cache: before it arrives, n DNS servers are visited, with round-trip times $RTT_1, \ldots, RTT_n$. Let RTT_0 be the round-trip time between your host and the Web server. Transmission times are negligible.

(a) The page is a single small HTML object. How long from the click until it is received?

(b) The HTML file now references eight very small objects on the same server. How long does it take with non-persistent HTTP and no parallel connections?

(c) With non-persistent HTTP and the browser set to open up to six connections in parallel?

(d) With persistent HTTP and pipelining?

What a cache is worth

E1.21: An institution has a 100 Mbps LAN, joined to the Internet by a 15 Mbps access link. Objects average 1 000 000 bits, and the institution's browsers request 16 per second. From the moment the router on the Internet side forwards a request, the answer takes 3 s on average to come back. Model the response time as the average access delay plus the average Internet delay. For the access delay use $\Delta/(1-\Delta\beta)$, where Δ is the time to send one object over the access link and β the arrival rate of objects.

(a) Find the total average response time. What does your number mean?

(b) A cache is installed in the LAN, with a miss rate of 0.4. Find the total response time again.

HTTP/2: frames and priorities

E1.22: A page consists of one video clip and five images. Over HTTP/2 the video travels as 2000 frames and each image as 3 frames. Count time in frame times.

(a) If all the video frames are sent first, without interleaving, how many frame times pass before all five images have been sent?

(b) If the frames are interleaved — one from each object in turn — how many?

(c) Now the images have priority over the video, and the first image over the second, the second over the third, and so on. How many frame times until the second image has been sent?

A conditional GET, typed by hand

E1.23: `printf` repeats its format for every argument, so each line below leaves with the `\r\n` that the protocol requires; the empty argument is the blank line that ends the header.

```
$ printf '%s\r\n' 'GET /hypertext/WWW/TheProject.html HTTP/1.1' \
                  'Host: info.cern.ch' 'Connection: close' '' \
  | nc info.cern.ch 80 | head -20
```

(a) Find the `Last-Modified:` line. Send a second request that adds an `If-Modified-Since:` header carrying that date, and obtain a 304 Not Modified. What is in the body this time?

(b) Which of the two replies would a cache prefer, and why?

(c) Send the conditional request again with an older date, say `Thu, 01 Jan 1970 00:00:00 GMT`. Why is the answer a 200 this time, and what does that tell you about how the server decides?

Persistent or not, on the wire

E1.24: Capture while fetching the same page twice in one command, first as it is, then asking the server to close after each reply:

```
$ u=http://info.cern.ch/hypertext/WWW/TheProject.html
$ curl -s -o /dev/null -o /dev/null $u $u
$ curl -s -H 'Connection: close' -o /dev/null -o /dev/null $u $u
```

Display filter `tcp.port == 80`.

(a) How many SYN segments does each command produce? Who sends the first FIN in the second case?

(b) Measure the RTT from the handshake. For each object, how many RTTs pass from the first packet of its connection — or from its GET — to the first byte of the reply?

(c) Do your numbers agree with the count made in the course for persistent and non-persistent connections?

Mail

E1.25 Alice sends a message from a Web-based mail account; Bob reads his mail with IMAP. List, in order, the application-layer protocols the message crosses from Alice's host to Bob's, and between which machines each one runs.

E1.26 How does SMTP mark the end of a message body? And HTTP? Could HTTP use SMTP's method? Why?

Who lied?

E1.27: Every mail server that handles a message adds a `Received:` line at the top of the headers, naming who handed the message over and who took it. These are the headers of a spam message, newest first:

```
Received: from barmail.cs.umass.edu ([128.119.240.3]) by cs.umass.edu; 7 Nov 2008 13:27:10
     -0500
Received: from asusus-4b96 (localhost [127.0.0.1]) by barmail.cs.umass.edu (Spam Firewall); 7
      Nov 2008 13:27:07 -0500
Received: from asusus-4b96 ([58.88.21.177]) by barmail.cs.umass.edu; 7 Nov 2008 13:27:07
     -0500
Received: from [58.88.21.177] by inbnd55.exchangeddd.com; 8 Nov 2008 01:27:07 +0700
From: "Jonny" <tennis5@pp33head.com>
To: <hg@cs.umass.edu>
```

(a) Only the originator is malicious; every other host is honest. Which host generated this message? Which lines can you trust?

(b) Look at the headers of a message you received recently. How many `Received:` lines are there, and can you find the IP address of the sender's host?

DNS: names and caches

E1.28 Can an organization's Web server and mail server have exactly the same alias, say `foo.com`? Which type of resource record holds the name of the mail server?

E1.29 Your department has a local DNS server for all its machines, and you are an ordinary user, not an administrator. Can you tell whether an external Web site was accessed from a department machine a few seconds ago? Explain how — or why not.

Down the tree by hand

E1.30: Do what your local server does when its cache is empty: start at a root server and follow the referrals down to the server that is authoritative for the name.

```
$ dig +norecurse @a.root-servers.net www.unina.it A      then ask the servers it names
```

(a) Write down the delegation chain: each server you asked, the zone it answers for, and the servers it sent you to next.

(b) Repeat for a name under a different top-level domain — a university in Japan, say.

(c) In the last answer, is the aa flag set, and what is the TTL? Ask your local resolver for the same name: which TTL is smaller, and why?

(d) Did any step answer with a CNAME? Where did the chain lead then?

Switch any VPN off first: a VPN may answer in place of the root server. Check the flags of every reply: a root server sets `aa` and never `ra`. If you see `ra`, somebody on the way is answering for it — which is the subject of L22.

Tools you need

- Wireshark: installed and set up for capture as a normal user, as shown with HTTP. Start the capture before the command.
- Terminal: `nc`, `curl`, `dig` and `ftp`. On Debian, Ubuntu and derivatives:

```
$ sudo apt install netcat-openbsd curl bind9-dnsutils tnftp
```

- Some networks block FTP or outgoing port 80. If an exercise cannot work where you are, try it from home and say so in class.
- Servers change: if a name or a page used here no longer answers, choose another one that does the same thing and keep the questions.